



Kishoreganj University
Department of Computer Science and Engineering
2nd Year 2nd Semester B.Sc. (Engg.) || Lab Final Examination-2023
CSE 2202: Data Structure and Algorithms II Lab (1.5 Credits)

Time: 3 Hours + 3 Hours

Full Marks: 40

Instructions

- Answer any one question from each section (from A and B) [3 hours].
- Section C is compulsory [3 hours].
- Each question in Section A and B carries 15 marks and Section C contains 10 marks.

SECTION-A

Question-A1

Toyota recently joined a ride-sharing company as a backend developer. His task is to develop a system that can keep track of different area IDs where the service of the company is available. Areas can become active or inactive at any time, so the system must be able to add, search, and remove area IDs quickly. To handle this, Toyota has been asked by his authority to use a skip list, which is good for fast insert, search, and delete operations. But since Toyota is a ChatGPT-engineer, he doesn't know how to make even a simple skip list. Now, he needs your help to build it. Your task is to help Toyota build this area management system using a menu-driven program that supports the following operations:

1. Insert an area ID
2. Search an area ID
3. Delete an area ID
4. Exit the program

Input

Each input line starts with an integer (1, 2, 3, or 4) denoting a menu option. If the number is 1, 2, or 3, it is followed by a space and an integer representing the area ID. Here, 1 means insert the area ID into the skip list, 2 means search for the area ID, and 3 means delete the area ID. If a line contains only the number 4, it means the program should exit.

Output

The output will show the result of each operation. For an insertion, it will print "Inserted: area ID <ID>". For a successful search, it will print "Found: area ID <ID>". If the area ID is not found during a search or deletion, it will print "Not Found: area ID <ID>". For a successful deletion, it will print "Deleted: area ID <ID>".

Sample Input Output

Input	Output
1 101	Inserted: area ID 101
1 203	Inserted: area ID 203
1 150	Inserted: area ID 150
2 150	Found: area ID 150
3 101	Deleted: area ID 101
2 101	Not Found: area ID 101
4	

Question-A2

Toyota has just landed in Erangel, his favorite map in PUBG. As soon as he lands, he starts collecting loot: guns, ammo, and healing items. Each loot item has a certain value (an integer), and Toyota wants to organize them in such a way that he can quickly access and manage them during the match. To do this, he decides to use an AVL Tree. You need to help Toyota for:

- Inserting each loot value (integer) into an AVL Tree.
- Print the loot values in sorted order along with their balance factors followed by the height of the tree.

Note: The AVL Tree should maintain its balancing property after every insertion.

Input

- The first line contains an integer n denoting the number of loot items.
- The next line contains n space-separated integers denoting the loot values collected by Toyota

Output

First, print the loot values in **descending** order. For each value in the tree, also print its balance factor on the same line. After that, print the height of the AVL tree on the next line. Note that, the balance factor of a node is calculated by the difference between the height of its left subtree and the height of its right subtree. Assume that the height counting starts from 1.

Sample Input Output

Input	Output	Explanation
5 10 20 30 40 35	Value: 40, Balance Factor: 0 Value: 35, Balance Factor: 0 Value: 30, Balance Factor: 0 Value: 20, Balance Factor: -1 Value: 10, Balance Factor: 0 Height: 3	

Question-B1

Toyota is a big fan of PUBG where even a fraction of a second can make the difference between survival and defeat. In each match, players try to quickly move between different locations to loot, take cover, or avoid danger zones. The map of the game is filled with interestingly named places like Pochinki, Rozhok, Georgopol, and many more. Usually, the game shows the real distances between these places. But today, the map is broken. The map no longer shows the real distances between places. Instead, the game now uses a different method. If two places are connected by a bidirectional road, the distance between them is calculated by adding the string-lengths of their names and then multiplying the result by 100. For example, if there are two places named Rozhok and Pochinki with a road between them, their name lengths are 6 and 8. So, the distance is: $(6 + 8) \times 100 = 1400$ meters. Now, Toyota has landed in one place (S) and wants to reach another (D). Your job is to help him find the shortest total distance he must travel on the map to travel from S to D .

Input

The first line of the input contains a single integer n . Next n line of input contains two strings X and Y each denoting there is a connecting both way road between two places: X and Y . The next line of input contains two strings S and D where S denotes the place where Toyota landed and D denotes the places where Toyota finally wants to go.

Output

Print the minimum distance from S to D . If D is unreachable from S then print -1.

Sample Input Output

Input	Output	Explanation
<pre>6 school pochinki rozhok georgopol rozhok school georgopol stalber school stalber stalber lipovka rozhok stalber</pre>	2500	

Question-B2

Toyota and his squad just landed into a new PUBG map. This map has several watchtowers, each placed at a specific (x, y) location. Players use these towers to hide from enemies. To move safely between the watchtowers, Toyota wants to set up hidden cables between them. He can connect any two watchtowers, but longer cables cost more. The cost to connect two watchtowers is calculated as:

$$|x_1 - x_2| + |y_1 - y_2|$$

Here, (x_1, y_1) is the position of the first watchtower and (x_2, y_2) is the position of the second watchtower. Now, Toyota wants to make sure all the watchtowers are connected, either directly or through other watchtowers. But he also wants to spend the least total cost to do this. Your task is to help Toyota find the minimum total cost needed to connect all the watchtowers.

Input

- The first line contains an integer n denoting the number of watchtowers.
- Next n lines contain two space separated integers x and y each denoting the coordinates of each watchtower.

Output

Print one number: the minimum total cost to connect all the watchtowers with hidden cables.

Sample Input Output

Input	Output	Explanation
<pre>4 2 -10 4 0 2 6 0 0</pre>	24	

Question-C1

Viva voce